THE REMIX

Covariates in DiD

Day 2 of 3: Afternoon

Scott Cunningham · Baylor University

University of Zurich · May 12, 2026

Causal Inference: The Remix · Yale University Press · forthcoming 2026

The PT-violation problem, take two

- This morning: a **third group** that shares the differential trend.
- But often, you don't have a clean placebo group.
- This afternoon: **condition on covariates that absorb the differential trend.**

Today's three estimators, and one false friend

OR (outcome regression) **IPW** (inverse propensity weighting) **DR** (doubly robust)

And: "*can't I just control for the propensity score in a regression?*"

That's the false friend. We'll see why before DR.

The applied problem

- Your treated states differ from your controls on observables — baseline income, demographics, industry mix, education.
- If $Y(0)$ trends differ across these dimensions, unconditional PT is implausible.
- But maybe *within* strata of X, PT holds:

$$\mathbb{E}[\Delta Y(0) \mid D = 1, X] \; = \; \mathbb{E}[\Delta Y(0) \mid D = 0, X]$$

- This is **conditional parallel trends**. The escape valve for designs without a placebo group.
- Three estimators target this. They differ in *which side* of the conditioning they model.

This afternoon, seven beats

1. Conditional parallel trends — the identifying assumption we'll lean on
2. OLS-with-controls: the bias decomposition that drives demand for everything below
3. Outcome regression (Heckman-Ichimura-Todd)
4. Inverse propensity weighting (Abadie 2005)
5. *The trap*: "can't I just control for the propensity score?"
6. Doubly robust DiD (Sant'Anna & Zhao 2020)
7. Hidden linearity bias — TWFE-with-controls fully exposed (Caetano & Callaway 2024)

We lead with the bias of OLS. Each fix below removes one piece of that bias.

Lesson

1. Conditional Parallel Trends

Conditional parallel trends, formally

The new identifying assumption

$$\mathbb{E}[\Delta Y(0) \mid D{=}1, X] \;=\; \mathbb{E}[\Delta Y(0) \mid D{=}0, X]$$

- Within strata of X, the untreated trend is the same for treated and control.
- Equivalently: *any* differential trend across treated and control is fully explained by their differences in X.
- Trade-off: weaker than unconditional PT (good), but you have to pick the right X (hard).
- Two ways to use X: model the outcome side ($\mu(X)$), or model the treatment side ($p(X)$). Each is one of our estimators.

Picking X: pre-treatment confounders, not post-treatment outcomes

- You want X that (i) predicts the treatment (causes selection) AND (ii) predicts $Y(0)$ (causes outcome differences).
- **Pre-treatment** only. Post-treatment X may be affected by D and will leak.
- Three practical screens (Imbens-Rubin, normalized differences):
  - Drop treated post-period, regress $Y(0)$ on candidate X
  - Within control units, regress $Y(0)$ on candidate X
  - Use $t-2$ and $t-1$ to estimate $\Delta Y(0)$ and regress on X

Source: Imbens & Rubin (2015); `02-Covariates.tex` frames 156–183.

Lesson

2. Outcome Regression

What OLS-with-controls gives you — and what it doesn't

Before any fix: what does OLS-with-controls actually identify?

The workhorse: $Y_{it} = \alpha_i + \gamma_t + \alpha\, D_{it} + \theta\, X_{it} + \varepsilon_{it}$. Estimate $\widehat{\alpha}$ by TWFE-OLS.

The Caetano-Callaway (2024) decomposition

$$\widehat{\alpha}_{\text{TWFE+}X} \xrightarrow{p} \underbrace{\overline{\text{ATT}}_w}_{\text{weighted ATT}} + \text{Bias}_A + \text{Bias}_B + \text{Bias}_C$$

- It is *not* the ATT. It is a *weighted average* of conditional ATTs — possibly with perverse weights — **plus** three bias terms.
- Each bias is a different assumption silently being made.
- **The bias is what creates the demand for OR, IPW, and DR.** They are each fixes for a piece of this expression.

The three bias terms, named

Bias A: heterogeneous τ in X

Treatment effect varies with X, but TWFE fits one θ. **Proof: next 4 slides (steps 1–4).**

Bias B: non-parallel X-trends across treated and control

Y^0 trends depend on X in different ways for treated vs. control. **Proof: 3 slides after that (steps 5–7).**

Bias C: the FE transformation drops X-levels and time-invariant Z

What you wrote as a control isn't what you ended up conditioning on. **Lesson 6.**

Three failures, three fixes. OR addresses A; IPW addresses A and B differently; DR closes both at once.

Outcome regression: model the trend, impute the counterfactual

Heckman, Ichimura & Todd (1997).

1. Use *only the control group* to fit a model of ΔY on X:

$$\widehat{\mu}_0(X) = \widehat{\mathbb{E}}[\Delta Y \mid D=0, X]$$

2. For each treated unit, predict the counterfactual trend $\widehat{\mu}_0(X_i)$.
3. The OR-DiD estimator:

$$\widehat{\delta}_{\text{OR}} = \frac{1}{N_1} \sum_{i:\, D_i=1} \big(\Delta Y_i - \widehat{\mu}_0(X_i)\big)$$

Interpretation: OR *imputes* what the treated would have done if untreated — using only the control group's X-to-trend mapping.

The TWFE-with-controls regression — and the three assumptions it hides

$$Y_{it} = \alpha_i + \gamma_t + \delta\,(T_t \times D_i) + \theta\,X_{it} + \varepsilon_{it}$$

- Standard move: add X_{it} to the TWFE. *"Sure! If you're willing to impose three more assumptions."*
- **(1) Additive functional form** in X (linear, no $D \times X$ interaction).
- **(2) Homogeneous treatment effects** in X (the slope θ doesn't depend on D).
- **(3) No X-specific trends differing across treated and control.**

Next slide: the proof of (2). Where does the homogeneity-in-X assumption come from? Trace it through the conditional expectations.

Proof, step 1 — the TWFE model and what conditional PT gives us

The TWFE model with one covariate X:

$$Y_{it} \;=\; \alpha_1 + \alpha_2\, T_t + \alpha_3\, D_i + \delta\,(T_t \times D_i) + \theta\, X_{it} + \varepsilon_{it}.$$

Conditional parallel trends says, for each value of X:

$$\mathbb{E}[Y_1^0 - Y_0^0 \mid D{=}1, X] \;=\; \mathbb{E}[Y_1^0 - Y_0^0 \mid D{=}0, X].$$

Rearrange. The missing counterfactual for the treated is identified:

$$\mathbb{E}[Y_1^0 \mid D{=}1, X] \;=\; \mathbb{E}[Y_0 \mid D{=}1, X] + \mathbb{E}[Y_1 \mid D{=}0, X] - \mathbb{E}[Y_0 \mid D{=}0, X].$$

Three observed conditional means $\Rightarrow$ one unobserved counterfactual. That's all PT does for us.

Proof, step 2 — plug TWFE in, the slopes look the same

From the TWFE model, the observed conditional means are:

$$\begin{aligned}\mathbb{E}[Y_0 \mid D{=}1, X] &= \alpha_1 + \alpha_3 + \theta\, X \\ \mathbb{E}[Y_1 \mid D{=}0, X] &= \alpha_1 + \alpha_2 + \theta\, X \\ \mathbb{E}[Y_0 \mid D{=}0, X] &= \alpha_1 + \theta\, X\end{aligned}$$

Plug into the identity from step 1:

$$\mathbb{E}[Y_1^0 \mid D{=}1, X] \;=\; \alpha_1 + \alpha_2 + \alpha_3 + \theta\, X.$$

The treated Y^1 post-period, from the TWFE model:

$$\mathbb{E}[Y_1^1 \mid D{=}1, X] \;=\; \alpha_1 + \alpha_2 + \alpha_3 + \delta + \theta\, X.$$

Both potential outcomes carry the same θ in X. Subtract: ATT $= \delta$. Clean. Suspiciously clean.

Proof, step 3 — what if Y^1 and Y^0 have different slopes in X?

Let the slopes differ. Write θ_1 for the Y^1 slope and θ_2 for the Y^0 slope:

$$\mathbb{E}[Y_1^1 \mid D{=}1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \delta + \boxed{\theta_1}\, X$$
$$\mathbb{E}[Y_1^0 \mid D{=}1, X] = \alpha_1 + \alpha_2 + \alpha_3 \qquad + \boxed{\theta_2}\, X$$

The true ATT given X:

$$\text{ATT}(X) \;=\; \mathbb{E}[Y^1 - Y^0 \mid D{=}1, X] \;=\; \delta \;+\; (\theta_1 - \theta_2)\, X.$$

- If $\theta_1 \neq \theta_2$, the treatment effect *depends on* X.
- TWFE fits **one** θ, not two.
- So $\widehat{\delta}$ recovers the ATT only when $\theta_1 = \theta_2$.

Proof, step 4 — the named assumption: *homogeneous* τ in X

Assumption

TWFE-with-X requires the treatment effect to be the same at every value of X.

- If X is **sex**: the effect is the same for men and for women.
- If X is **income** (continuous): the effect is the same at $1 and at $1,000,000.
- If X is **age**: the effect is the same at 25 and at 65.

This is a structural assumption on the DGP. *OLS doesn't tell you you're making it. The math does.*

If the effect actually varies with X, TWFE-with-X averages across the variation and gives you a weighted parameter you didn't ask for.

Proof, step 5 — the second restriction starts with four cell means

Assume now the homogeneity in X holds, so a single θ governs both potential outcomes. Take expectations of the TWFE equation at each of the four cells:

$$
\begin{aligned}
\mathbb{E}[Y_1 \mid D{=}1] &= \alpha_1 + \alpha_2 + \alpha_3 + \delta + \theta\, X_{11} \\
\mathbb{E}[Y_0 \mid D{=}1] &= \alpha_1 + \phantom{\alpha_2 +{}} \alpha_3 \phantom{{}+ \delta} + \theta\, X_{10} \\
\mathbb{E}[Y_1 \mid D{=}0] &= \alpha_1 + \alpha_2 \phantom{{}+ \alpha_3 + \delta} + \theta\, X_{01} \\
\mathbb{E}[Y_0 \mid D{=}0] &= \alpha_1 \phantom{{}+ \alpha_2 + \alpha_3 + \delta} + \theta\, X_{00}
\end{aligned}
$$

$X_{dt} = \mathbb{E}[X \mid D{=}d, T{=}t]$: the cell-mean of X. Four cells, four X-means.

Proof, step 6 — take the DiD, what survives?

Form the DiD on cell means. The α's cancel, δ survives, and a bracketed term in X-cells is left:

$$\delta^{\mathsf{DD}} = \delta + \theta\left[(X_{11} - X_{10}) - (X_{01} - X_{00})\right].$$

For δ^{DD} to equal the true ATT δ, the bracket must vanish:

$$\boxed{(X_{11} - X_{10}) = (X_{01} - X_{00})}$$

The X trend for the treated must equal the X trend for the controls.

Proof, step 7 — the assumption: Y^0 trends don't depend on X

Assumption

The evolution of the untreated potential outcome Y^0 over time is the same regardless of X.

- Rich people cannot be on a different trend than poor people.
- Men cannot be on a different trend than women.
- Old cannot be on a different trend than young.

This is a restriction on the DGP, not on the data. *Two assumptions, named:* ***(homogeneity)*** *τ doesn't depend on X.* ***(parallel X-trends)*** *Y^0 trends don't depend on X. Without both, TWFE-with-X does not identify the ATT.*

Why OR can fail: it's all model

- OR's entire identification rests on $\widehat{\mu}_0(X)$ being correctly specified.
- Get the functional form wrong — a missing nonlinearity, a missing interaction — and OR is biased.
- Especially fragile in the post-period, where you're *extrapolating* from the control distribution of X to the treated distribution.
- If treated units have X-values rare among controls, OR is leaning on the model where the data are thin.

This is why IPW exists — as a different way to use X.

LESSON

3. Inverse Propensity Weighting

The propensity score, in one minute

- Definition: $p(X) = \Pr(D{=}1 \mid X)$ — the probability of treatment given covariates.
- Estimated by logit/probit **maximum likelihood**:

$$\widehat{\theta} = \arg\max_{\theta} \sum_{i} \left[D_i \log p(X_i;\theta) + (1 - D_i) \log(1 - p(X_i;\theta)) \right]$$

- **Rosenbaum-Rubin (1983)**: if D is unconfounded given X (a vector), it is unconfounded given $p(X)$ (a scalar).
  One number summarizes all of X.
- The pscore lives in $[0, 1]$. Treated units have higher $p(X)$ on average. The pscore reorganizes the data along the dimension of selection.

The Rosenbaum-Rubin theorem (1983)

Statement

If treatment is unconfounded given X, i.e. $\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$,
then it is also unconfounded given the propensity score $p(X)$ alone:

$$\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid p(X).$$

- X is a vector. $p(X)$ is a scalar in $[0, 1]$. **One number compresses all the confounding information.**
- Operationally: matching, weighting, or balancing on $p(X)$ gives the same identification as on the full X.
- This is what licenses IPW. It does *not* license putting $p(X)$ on the right-hand side of a regression — that's a different use, with different consequences (Lesson 4 trap).

Abadie (2005): reweight the controls

- Idea: reweight control units by $\dfrac{p(X)}{1-p(X)}$ so the reweighted control sample *looks like* the treated sample in X.
- Then take a simple DiD on the reweighted sample.

The Abadie IPW-DiD estimator

$$\widehat{\delta}_{\text{IPW}} = \frac{1}{N_1}\sum_i \left(\frac{D_i - p(X_i)}{1-p(X_i)}\right)\Delta Y_i$$

The operator $(D-p)/(1-p)$ gives weight $+1$ to treated units and weight $-p/(1-p)$ to controls — exactly the reweighting in line one. **Identification:** under conditional PT + common support, IPW recovers the ATT. Proof next slide.

Proof: IPW recovers the ATT under conditional PT

1. The operator we need to evaluate:

$$\mathbb{E}\left[\frac{D - p(X)}{1 - p(X)}\,\Delta Y\right] = \mathbb{E}\left[\frac{D\,\Delta Y}{1 - p(X)}\right] - \mathbb{E}\left[\frac{p(X)\,\Delta Y}{1 - p(X)}\right].$$

2. Iterate expectations over X, then condition on D. For each piece:

$$\mathbb{E}\left[\tfrac{D\,\Delta Y}{1-p(X)} \mid X\right] = \tfrac{p(X)}{1-p(X)}\,\mathbb{E}[\Delta Y \mid D{=}1, X]$$

$$\mathbb{E}\left[\tfrac{p(X)\,\Delta Y}{1-p(X)} \mid X\right] = \tfrac{p(X)}{1-p(X)}\,(1 - p(X))\,\mathbb{E}[\Delta Y \mid D{=}0, X] \cdot \tfrac{1}{1-p(X)} = \tfrac{p(X)}{1-p(X)}\,\mathbb{E}[\Delta Y \mid D{=}0, X]$$

3. Subtract:

$$\mathbb{E}\left[\tfrac{D-p(X)}{1-p(X)}\Delta Y \,\middle|\, X\right] = \tfrac{p(X)}{1-p(X)}\,\big\{\mathbb{E}[\Delta Y \mid 1, X] - \mathbb{E}[\Delta Y \mid 0, X]\big\}.$$

4. Apply conditional PT. The bracket equals $\mathbb{E}[\Delta Y(1) - \Delta Y(0) \mid D{=}1, X] = \text{ATT}(X)$:

$$= \tfrac{p(X)}{1-p(X)}\,\text{ATT}(X).$$

5. Take expectations over X and normalize by N_1/N. After cancellation,
$\widehat{\delta}_{\text{IPW}} \to \mathbb{E}[\text{ATT}(X) \mid D{=}1] = \text{ATT}.$ ■

Common support is non-negotiable

- IPW's weights are $\dfrac{p(X)}{1-p(X)}$ for controls. As $p(X) \to 1$, the weight $\to \infty$.
- If some treated X-values have $p(X) \approx 1$, then NO control unit is comparable — and the weights are unstable.
- Standard responses:
  - Trim units with $p(X) > 0.95$ (or < 0.05).
  - Report sensitivity to the trim threshold.
  - In the limit, "weak overlap" breaks identification entirely. (Advanced deck.)

The pscore optimizer (IRLS, Newton-Raphson) can also misbehave near separation. Forward link to advanced.

LESSON

4. The Trap

"Can't I just control for the pscore?"

A natural temptation

> *If the propensity score is the one-scalar summary of confounding, why not just add it as a regressor? Done.*

— The audience, right now

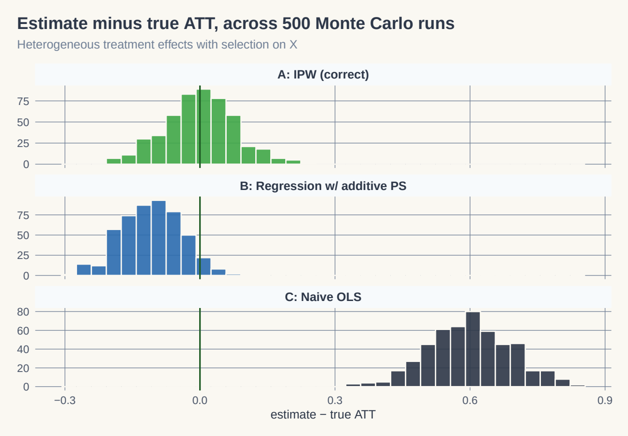
- It sounds right. Rosenbaum-Rubin says *one scalar suffices*. Regression handles scalars.
- So: $Y_i = \alpha + \tau D_i + \gamma \widehat{p}(X_i) + \varepsilon_i$
- Surely this recovers τ as the ATT? **No.** And the reason matters.

What that regression actually assumes

$$Y_i = \alpha + \tau D_i + \gamma\,\widehat{p}(X_i) + \varepsilon_i$$

- No $D \times X$ interaction. No $D \times \widehat{p}$ interaction.
  The treatment effect is forced constant across the entire range of $\widehat{p}(X)$.
- Equivalently: the same dose of policy produces the same effect on a person with $\widehat{p}(X) = 0.10$ as on a person with $\widehat{p}(X) = 0.90$.
- That's an assumption about **the outcome side** — one that Rosenbaum-Rubin says nothing about.
- Rosenbaum-Rubin gives you a valid *reweighting* tool. It does not authorize regression-on-pscore.

A simulation makes it visible



*500 reps. Heterogeneous $\tau(X) = 1 + 0.5\,X_1$. Method A (IPW reweighting): bias ≈ 0. Method B (regression with $\widehat{p}$): **bias** $\approx -10\%$.*

The simulation, one screen of R

```
N <- 1000
X1 <- rnorm(N); X2 <- rnorm(N)
p <- plogis(0.5*X1 + 0.3*X2 - 0.2) # true propensity
D <- rbinom(N, 1, p)
Y0 <- X1 + 0.5*X2 + rnorm(N)
Y1 <- Y0 + (1 + 0.5*X1) # heterogeneous tau(X)
Y <- D*Y1 + (1-D)*Y0

phat <- glm(D ~ X1 + X2, family=binomial)$fitted.values

# Method A: IPW -- reweight controls
w <- ifelse(D==1, 1, phat/(1-phat))
A <- mean(Y[D==1]) - weighted.mean(Y[D==0], w[D==0])

# Method B: regression with pscore as a control
B <- coef(lm(Y ~ D + phat))["D"] # <-- the trap
```

Full simulation: `decks/code/IPW-Demo/ipw_simulation.R`

Weight versus regressor — different uses of the same scalar

- The propensity score is a **tool**. The same scalar can be used as:
  - a **weight** (Method A: reweight controls) — recovers ATT
  - a **regressor** (Method B: add as control) — imposes homogeneous effects
- Same number. Different uses. *Different identifying assumptions.*
- Method B's bias is not noise. It is the heterogeneity in $\tau(X)$ that the regression cannot see.

The lesson before DR

"Doubly robust" is sometimes mis-summarized as "IPW plus a regression control."

That's wrong. DR is something else — it uses the pscore as a weight, AND it uses an outcome model.
But it does NOT use the pscore as a regressor.

Name the disease: ***additive-control-side bias***

Whenever you put a covariate on the right-hand side of an outcome regression *additively* (no $D \times X$ interaction), you force:

τ is constant across all values of that covariate.

- Method B (Lesson 4): the covariate is $\widehat{p}(X)$ — a scalar summary of X.
  Force constant τ across $\widehat{p}(X)$. Heterogeneity in $\tau(X)$ is invisible.
- TWFE-with-controls (Lesson 2): the covariate is X itself.
  Force constant τ across X. Heterogeneity in $\tau(X)$ is invisible.
- **Same disease, different costume.** Caetano & Callaway (2024) name the TWFE version *hidden linearity bias*. It's the same root.

DR's escape: use X and $\widehat{p}(X)$ as weights and to model the counterfactual $\mu_0(X)$ on controls only — never as additive regressors of Y.

LESSON

5. Doubly Robust DiD

Doubly robust: two chances to be right

- OR needs $\widehat{\mu}_0(X)$ correctly specified.
- IPW needs $\widehat{p}(X)$ correctly specified.
- Neither is foolproof. Each fails if its own model is wrong.

The Sant'Anna & Zhao (2020) move

Combine them. The **DR-DiD** estimator is consistent if *either* model is correct — not necessarily both.

Two chances to be right. That is what "doubly robust" means.

DR-DiD identification: three assumptions

Assumption 1 (Data)

Panel or repeated cross-sections with pre- and post-periods; treatment in post-period; covariates X pre-treatment.

Assumption 2 (Conditional parallel trends)

$$\mathbb{E}[\Delta Y(0) \mid D{=}1, X] \;=\; \mathbb{E}[\Delta Y(0) \mid D{=}0, X]$$

Assumption 3 (Common support)

$$0 \;<\; p(X) \;<\; 1 \text{ for almost every } X.$$

DR-DiD + these three $\Rightarrow$ ATT identified if either model is correct. Proof next slide.

The DR-DiD estimator

$$\widehat{\delta}_{\mathsf{DR}} = \mathbb{E}\left[\left(\frac{D}{\mathbb{E}[D]} - \frac{\frac{p(X)(1-D)}{1-p(X)}}{\mathbb{E}\left[\frac{p(X)(1-D)}{1-p(X)}\right]} \right) (\Delta Y - \mu_0(X)) \right]$$

- $p(X)$: propensity score model
- $\mu_0(X)$: outcome model fitted on *controls*: $\mu_0(X) = \mathbb{E}[\Delta Y \mid D{=}0, X]$
- The IPW weight reweights the comparison. The OR term centers the residual on the imputed counterfactual.
- If p is right, the IPW piece does the work. If μ_0 is right, the OR piece does the work. Proof next slide.

Proof: doubly robust — *either* model can be wrong

Let $w(X)$ denote the weight operator in $\widehat{\delta}_{\text{DR}} = \mathbb{E}[w(X)(\Delta Y - \mu_0(X))]$.

Case A: $p(X)$ **correct,** $\mu_0(X)$ **may be wrong.**

$$\widehat{\delta}_{\text{DR}} \;=\; \underbrace{\mathbb{E}[w(X)\,\Delta Y]}_{\text{IPW identifies ATT}} \;-\; \underbrace{\mathbb{E}[w(X)\,\mu_0(X)]}_{=\,0 \text{ in expectation}} \;=\; \text{ATT}.$$

The OR term vanishes because $\mathbb{E}[w(X) \mid X] = 0$ once $p(X)$ is correct. **IPW alone identifies the ATT.**

Case B: $\mu_0(X)$ **correct,** $p(X)$ **may be wrong.** Under conditional PT, the residual $\Delta Y - \mu_0(X)$ has conditional mean zero given $D{=}0, X$:

$$\mathbb{E}[\Delta Y - \mu_0(X) \mid D{=}0, X] \;=\; 0,$$

so the control piece zeroes regardless of weighting. The treated piece $\frac{D}{\mathbb{E}[D]}(\Delta Y - \mu_0(X))$ integrates directly to the ATT. **OR alone identifies the ATT.**

Either suffices — not both. That is "two chances to be right." ■

DR is *not* regression-with-pscore

- Look at the formula. $\widehat{p}(X)$ **appears as a weight, never as a regressor.**
- $\mu_0(X)$ models the outcome, but on *controls only* — to impute the counterfactual trend.
- The two pieces are combined by a re-centering and reweighting, NOT by adding $\widehat{p}$ to a regression's right-hand side.

The pscore is the tool. ***Using it as a weight is doubly robust.*** *Using it as a regressor is the trap from Lesson 4.*

Efficiency and the influence function

- Under conditional PT + common support, DR-DiD is **semiparametrically efficient**.
- No estimator of the ATT that uses only the same assumptions can have a smaller asymptotic variance.
- The asymptotic variance has a closed form via the *influence function* (this is what the `drdid` package computes for SEs).

Practical recipe

Pick X for confounding (Lesson 1). Estimate $\widehat{p}(X)$ by logit. Estimate $\widehat{\mu}_0(X)$ on controls. Plug into the DR-DiD formula. Influence-function SEs.

DR-DiD, in code

```r
library(DRDID)

# Panel data: outcome y, treatment indicator d, time t, id, covariates X
result <- drdid(
  yname = "y",
  tname = "t",
  idname = "id",
  dname = "d",
  xformla = ~ x1 + x2 + x3, # covariates for BOTH p(X) and mu_0(X)
  data = panel_df,
  estMethod = "imp", # Sant'Anna & Zhao (2020) DR
  weightsname = NULL,
  boot = FALSE,
  inffunc = TRUE # influence-function SEs
)
summary(result)
```

Same X used for both models. Misspecify one — the other catches you.

LESSON

6. Hidden Linearity Bias

Caetano & Callaway 2024

TWFE-with-controls has two distinct problems

The workhorse specification $Y_{it} = \alpha_i + \gamma_t + \tau D_{it} + X_{it}'\beta + \varepsilon_{it}$ is biased for two *separate* reasons:

1. **The FE transformation throws away what you wanted to control for.**
   First-differencing X_{it} leaves only ΔX. The *level* of X is gone. Time-invariant Z is gone entirely. (Caetano-Callaway 2024 — this lesson.)
2. **What's left enters additively, forcing constant τ.**
   Whatever covariate survives the FE transformation sits on the RHS as an additive regressor with no $D \times X$ interaction. Same disease as Method B (Lesson 4).

This lesson covers problem (1). Problem (2) is the L4 callback we'll close on.

An applied problem you've already run

> *TWFE with time-varying controls is the workhorse DiD specification.*
> *Almost every DiD paper runs it.*

— Anybody who's seen a job-market paper

- Standard recipe: include population, income, demographic shares, region indicators, etc. on the right-hand side. Run TWFE. Done.
- **What if I told you that regression doesn't control for what you think it controls for?**
- Caetano & Callaway (2024, R&R *REStat*): yes. The fixed-effect transformation quietly drops your X levels and your time-invariant Z.

An application: stand-your-ground laws (Cheng & Hoekstra 2013)

- 50 states, 2000–2010. 20 states adopt stand-your-ground laws. Outcome: log homicide rate.
- Treated states differ from controls in **levels**:
  - South dummy: std. diff. $= 0.81$ (large)
  - Poverty rate: std. diff. $= 1.06$ (huge)
  - Median income: std. diff. $= -1.08$ (huge)
- But the differences in **changes** of these covariates are small.

The headline result

TWFE with covariates: $\widehat{\alpha} = +0.067$, significant.
AIPW with the right covariates: $\widehat{\text{ATT}} \approx 0$, insignificant.

The positive TWFE result is largely artefact of hidden linearity bias.

What TWFE actually controls for, after the FE transformation

You wrote down this model:

$$Y_{it} = \alpha_i + \gamma_t + \tau D_{it} + X_{it}'\beta + Z_i'\delta + \varepsilon_{it}$$

First-difference (or within-transform) to kill α_i:

$$\Delta Y_{it} = \widetilde{\gamma}_t + \tau\,\Delta D_{it} + \Delta X_{it}'\beta + \Delta\varepsilon_{it}$$

- α_i vanishes (good).
- Z_i vanishes too (time-invariant). **Region: gone.**
- X_{it} becomes ΔX_{it}. **Population level: gone. Only population growth remains.**
- Your "controls" are now *changes* in X, not levels.

What you should be controlling for

The correct first-differenced model for $\Delta Y(0)$ (Caetano-Callaway Eq. 4):

$$\Delta Y_{it}(0) = \widetilde{\gamma}_t + \underbrace{Z_i'\widetilde{\delta}_t}_{\text{time-inv}} + \underbrace{\Delta X_{it}'\beta_t}_{\text{changes}} + \underbrace{X_{i,t-1}'\widetilde{\beta}_t}_{\text{level}} + \Delta\varepsilon_{it}$$

- **Time-invariant** Z (region) belongs on the right — because trends differ by region.
- **Pre-period level** $X_{i,t-1}$ belongs on the right — because the level of population predicts the trend in homicides.
- TWFE drops both. **Hidden linearity bias.**

The L4 callback: problem (2) is the additive-RHS disease

We've covered problem (1) above — the FE transformation throws away levels. Problem (2) is the L4 disease, present in TWFE-with-controls too.

Estimator	**What goes on RHS additively**	**What gets forced constant**
Method B (L4)	$\widehat{p}(X)$ as control	τ across $\widehat{p}(X)$
TWFE-with-controls (L2, L6)	ΔX as control	τ across ΔX
General formulation	$W = f(X)$ as additive RHS regressor	τ across W

- Problem (1) is C&C's contribution: *what* you control for is wrong.
- Problem (2) is the L4 disease: *how* you control for it is wrong.
- TWFE-with-controls catches both. DR escapes both — using X and $\widehat{p}(X)$ as **weights**, not as additive regressors.

The fix: AIPW with $(X_{t-1},\ Z)$

Caetano-Callaway's recommended estimator:

$$\widehat{\text{ATT}} = \widehat{\mathbb{E}}_{D=1}[\Delta Y - \widehat{m}_0(X_{t-1}, Z)]$$

- $\widehat{m}_0(X_{t-1}, Z)$: outcome regression of ΔY on *pre-period level* of X and *time-invariant* Z — fit on **controls only**.
- Combine with IPW on a generalized propensity score (also using X_{t-1}, Z) for full doubly-robust safety.
- R packages: `pte`, `twfeweights`.
- This is just DR-DiD (L5) with the *right* conditioning set: levels of time-varying X + time-invariant Z.

Stand-your-ground revisited — the numbers

Specification	$\widehat{\text{ATT}}$	SE
TWFE with ΔX (the standard recipe)	$+0.067$	0.026
Reg. adj. with $\Delta X + Z$	$+0.106$	0.058
Reg. adj. with $X_{t-1} + Z$ (*Caetano-Callaway preferred*)	$+0.019$	0.043
Reg. adj. with $\Delta X + X_{t-1} + Z$	$+0.078$	0.180

- The TWFE positive effect ($+6.7\%$) shrinks to near-zero ($+1.9\%$) when you control for *levels* of X and Z.
- The bias was hidden in plain sight — the regression compiled fine. It just wasn't controlling for what we thought.

LESSON

7. Compositional Change

Two flavors of “X moves”

Flavor	What’s moving	Estimator
Lesson 6	Same units, X_{it} moves over time	Caetano-Callaway AIPW
Lesson 7	New units each period, X distribution moves	Hong / Sant’Anna-Xu

- Both break unconditional PT *in finite samples* even when it holds in the population.
- L6 (panel): the FE transformation loses the X level. L7 (repeated cross-sections): the sample composition shifts.
- Different fixes. Same family of concerns: X **is moving, and your estimator pretends it isn’t.**

Hong (2013): Napster, internet users, and music expenditure

- Repeated cross-sections, 1996 and 2002.
- In 1996, internet users were a small, atypical slice — mostly young, urban, technical.
- By 2002, internet users looked much more like the general population.
- Compare “internet users” to “non-users,” 1996 vs. 2002.
  Did Napster reduce music spending — or did the composition of internet users change?
- Hong’s answer: the naive DiD confounds the two. The compositional shift accounts for much of the apparent “Napster effect.”

Hong (2013) JEEA. Figures in `CI-2/lecture_includes/hong_napster.png`.

So why do we need Sant'Anna & Xu given Hong showed it?

Hong (2013) gave us the intuition — and a single empirical example. To use this in your own work you need three things Hong didn't formalize:

1. **A target estimand.** What *is* “the” ATT under compositional shift? Is it the average over the post-period sample (your data) or the pre-period sample? These are different parameters.
2. **A test.** Hong showed his data had the problem. How do *you* know yours does, without the benefit of a second empirical paper?
3. **A correction.** If your test rejects, what estimator should you report? Hong didn't give you one — he flagged the problem.

The story stays **Hong’s.**

The toolkit is **new.**

Sant'Anna & Xu: first, fix an estimand

> *What is "the" ATT when the sample composition is moving?*

— Anyone trying to use Hong in their own paper

The SX target

The population ATT averaged over a **fixed reference X-distribution** — typically the pre-period.

- The naive DiD targets a *moving* mixture — a weighted average of period-specific ATTs, weighted by whatever happened to be in each period's sample.
- Pick a reference. Now you know what number you're trying to estimate.

Second, test for compositional shift

Hausman-style test

Compute the naive DiD. Compute the DiD on the sample *reweighted to the reference X-distribution*. Compare.

- Under the null (no compositional shift), the two estimates agree.
- Under the alternative, they diverge — and the gap is informative about how much your point estimate is doing compositional vs. behavioral work.
- Standard χ^2 inference.

This is the move Hong didn't have. He showed his data had the problem. The Hausman test lets you check yours.

Third, reweight and estimate

The SX estimator

Reweight units in each period so the within-period X-distribution matches the chosen reference. Then compute DiD on the reweighted sample.

- Mechanically: IPW-style reweighting, with weights derived from $f_{\text{ref}}(X)/f_t(X)$ at each period.
- **Doubly robust** version available — combine the IPW reweighting with an outcome model (callback to Lesson 5).
- Software: implementable from the IPW machinery you already know.

When compositional change matters in practice

- **Survey data over long horizons.** CPS, NLSY, ACS — sample composition drifts as response rates, demographics, frame definitions change.
- **Administrative outcomes with selective coverage.** Who enters the data may itself respond to policy (Medicaid take-up, charter-school enrollment).
- **New technology adoption.** Hong's Napster, again. The adopter pool changes over time — almost by definition.
- **Long pre-periods in event studies.** The further back you go, the more likely the population you're comparing has shifted.

First defense: check whether X distribution moved across time. Then run the Hausman test. Then report both estimates.

Lab: LaLonde under selection (*the four estimators, one truth*)

- **The benchmark.** The experimental NSW sample identifies the ATT by random assignment: $\widehat{\text{ATT}}^{\text{RCT}} \approx \$1{,}700$ **(1978 dollars)**. Compute it first.
- **The non-experimental panel.** NSW treated $+$ a random CPS sample as the control. Selection bias is large and *negative* — naive DiD gives the wrong sign.
- **Run all four estimators**, get an ATT, compare to \$1,700:
  - **TWFE-with-X** (interact Post $\times$ X — without the interaction, FE sweeps them out)
  - **OR** (Heckman–Ichimura–Todd 1997)
  - **IPW** (Abadie 2005)
  - **DR-DiD** (Sant'Anna–Zhao 2020): `drdid` with the `all` option
- **Then answer:** which estimators move the ATT in the right direction? Which get closest? When does DR *not* dominate? What does the propensity-score overlap plot show?

Walkthrough: `zurich/labs/Lalonde-Covariates/README.md`. Data: `lalonde_exp_panel.dta` & `lalonde_nonexp_panel.dta` on Mixtape-Sessions GitHub.

What we've done this afternoon

1. Conditional PT — the new identifying assumption
2. Outcome regression: model the trend, impute the counterfactual
3. IPW: reweight controls to look like the treated
4. The trap: "controlling for the pscore" imposes constant τ across $\widehat{p}(X)$
5. Doubly robust: pscore as a *weight* (not regressor) + outcome model
6. Hidden linearity bias (Caetano-Callaway): *same disease*, costume = TWFE-with-controls
7. Compositional change (Hong; Sant'Anna-Xu): when the *units* shift, not X

The unifying lesson: covariates on the additive RHS of an outcome regression impose constraints on τ.
*DR's escape is to use them as **weights**, not regressors.*

Tomorrow

- **Day 3:** when even covariates can't save you. Synthetic control, the frontier methods.
- Plus: when treatment timing differs across units — the staggered-adoption mess (Goodman-Bacon, Callaway-Sant'Anna, Sun-Abraham, BJS, dCdH).

Use the pscore as a **weight,**

not as a **regressor.**

That is the difference between **doubly robust** and **doubly biased.**